Original Article

The Optical and Photoluminescence Features of GdNiSe Material Synthesized Hydrothermally

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ABSTRACT

In this study, the hydrothermal method was used to synthesize the gadolinium nickel selenide (GdNiSe) material. The materials' energy bandgaps ranged from 1.58 eV to 1.18 eV. Annealing the material led to a reduction in the energy band gap. The energy bandgap of GdNiSe was enhanced through the annealing process . The photoluminescence was carried out by exposing the materials to sunlight at different time intervals. The unannealed GdNiSe material was exposed to sunlight at 20-minute intervals, ranging from 0 to 140 minutes. The prominent peaks were identified within the wavelength range of 612 to 661 nm. X-ray diffraction (XRD) analysis of the GdNiSe material has shown that it crystallizes in a hexagonal crystal structure with space group P63/mmc. The lattice parameters of GdNiSe are a = 0.5875 nm and c = 1.0256 nm. The XRD patterns of GdNiSe material are affected by annealing. It impacts the intensity, position, and size of peaks, revealing details about the material's structure and microstructure. The FTIR spectra showed various peaks, such as 2,654, 1,869, 1,638, 1,105, and 935 cm^{-1}. The GdNiSe material exhibits a strong covalent bond, leading to a short bond length and high bond strength. The overlapping of their 3d and 4p orbitals creates the bond between Ni and Se, resulting in a bond order of two.

Introduction

The versatile characteristics of transition metal chalcogenides have led to significant interest in their potential for nanotechnology and various applications, including energy and biomedical applications [1-4]. Energy plays a central role in every economy, as technological advancements rely on it. Energy consumption has increased consistently as the world becomes more advanced [5–8]. Currently, there is a

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growing interest in affordable photovoltaic cells compared with expensive fossil fuels. The rise in petroleum prices results from increased demand driven by population growth, creating upward pressure. The photovoltaic industry primarily uses silicon-based solar cells. Direct bandgap materials have an advantage over silicon in their use as absorber materials due to their direct bandgap. Lower energy requirements and costs give thin film solar cells an advantage over silicon solar cells, allowing for large-scale production [9–11]. The atomic-scale thickness and ideal bandgap structures of two-dimensional materials make them highly promising for applications in nanoelectronics, optoelectronics, and energy conversion and storage. The focus of recent research has been on 2D layered materials, primarily due to their strong covalent or ionic bonding within in-plane atoms, weak van der Waals forces between interlayers, and exceptional electrical, optical, and mechanical properties resulting from their smooth surface and absence of chemical dangling bonds [7].

The optical and electronic properties of semiconductor materials based on nickel selenide are exceptional, which is why they are highly sought after in materials science applications. Their wide bandgap, wider than bulk nickel selenide, makes these semiconductors ideal for catalysis, laser materials, and energy storage devices. The utilization of nickel chalcogenides in electronic devices is becoming more popular. Nickel selenide is seldom used in electronic devices. The p-type semiconductor NiSe has attracted considerable interest for its wide range of applications in electronic devices [12-15]. The purpose of this study was to synthesize GdNiSe and analyze its optical and photoluminescence emission (PL) properties for a range of applications.

Experimental

In the synthesis of GdNiSe, the gadolinium (III) nitrate hexahydrate $Gd(NO_3)3.6H_2O$ (0.1M), nickel (II) nitrate hexahydrate. $Ni(NO_3)_2{\cdot}6H_2O$ (0.1M), selenium dioxide SeO_2 (0.1M), and triethanolamine TEA (2 mL) were dissolved in (20 mL) of distilled water. A solution of triethanolamine TEA was prepared to act as a precipitant. While stirring continuously, the nickel(II) nitrate hexahydrate. $Ni(NO_3)_2{\cdot}6H_2O$ solution and the selenium dioxide SeO_2 solution were combined. After 30 min, stoichiometric amounts of gadolinium (III) nitrate hexahydrate $Gd(NO_3)_3.6H_2O$ was added to the solution. After stirring this solution for 5 hours, it was transferred to a Teflon lined autoclave and treated it at 150 °C for 18 hours, see Figure 1. After cooling in the ambient air, the precipitate from the autoclave was washed with distilled water, and then filtered. The dehydrated powder underwent drying at 180 °C. The GdNiSe nanoparticles were further annealed at different temperatures (220, 240, and 260 °C).

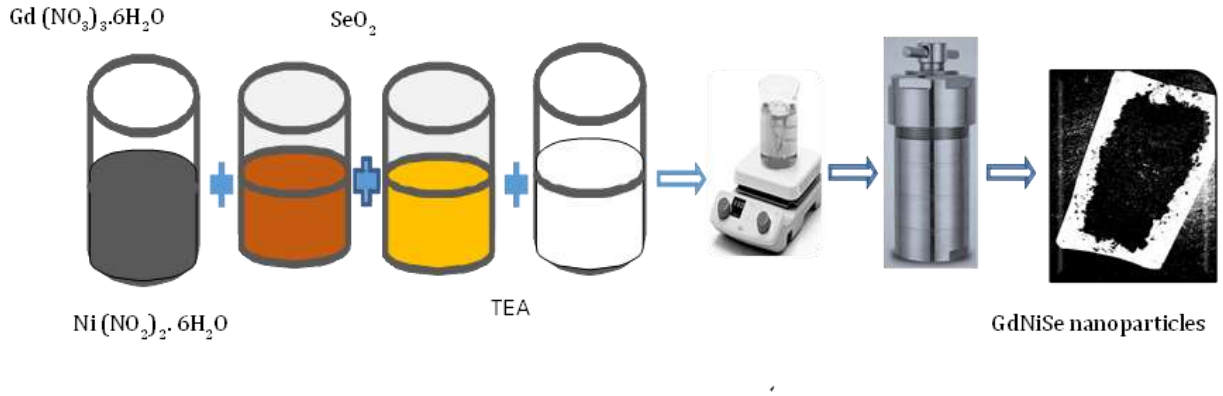


Figure 1: Schematic diagram of the synthesis of GdNiSe material

Materials for Characterization

Certain analytical tools were used to thoroughly characterize the synthesized materials. A UV-VisIS-NIR 756S spectrophotometer was used to measure absorbance (300–800 nm), which made it possible to calculate different optical parameters for each sample. A PANalytical X'Pert PRO MPD (45 kV, 40 mA) and XRD was used for the structural characterization of the films, XRD analysis (validated the crystal structure and peak intensities Cu-Kα, α = 1.5406 Å; 0.5°–130° 2, and 0.5 sec/step), and database software (International Centre for Diffraction Data). The GdNiSe material was examined for their PL using a spectrofluorometer (Perkin Elmer LS-45).

Results

Optical Study of GdNiSe Material

Figure 2a depicts the absorbance of GdNiSe material synthesized with a hydrothermal approach. The absorbance shows a decrease that aligns with the wavelength of the light radiation, exhibiting prominent peaks in the visible region of the spectra. Increasing the annealing temperature improved the absorbance of the synthesized GdNiSe material. The material's absorbance rises with higher annealing temperatures, enhancing the photoluminescence properties of the synthesized material. The electronic structure of the hydrothermally synthesized GdNiSe material is responsible for its absorbance. The bandgap of GdNiSe is approximately 1.58 eV, making it a narrow-bandgap semiconductor. The excitation of electrons from the valence band to the conduction band causes the absorption of light in the visible and near-infrared regions. Annealing the GdNiSe material improves its electromagnetic and dielectric properties, resulting in higher AC conductivity and dielectric constant compared to the unannealed GdNiSe. Improved absorption and optical properties are achieved by annealing the material. The structural and optical properties of GdNiSe materials were improved through annealing. Annealing the material caused a slight shift of the absorption edge toward lower wavelengths. Varying the annealing temperature from 220 to 260 °C causes changes in the absorption spectra of GdNiSe material. With increasing annealing temperature, the material's absorption edge moves to lower energies, suggesting a narrow bandgap. The annealing process caused structural changes and improved the crystallinity of the GdNiSe layer. The optical properties of GdNiSe depend on the annealing temperature. Increasing annealing temperatures improves crystallinity and decreases defects, altering the material's absorption spectrum. Optoelectronic devices like solar cells and photodetectors can benefit from using GdNiSe material. GdNiSe's strong absorbance in the visible and near-infrared regions makes it ideal for light-harvesting applications. Figure 2b depicts the transmittance of GdNiSe material. The transmittance shows an increase that aligns with the wavelength of the light radiation, exhibiting prominent peaks in the visible region of the spectra. Increasing the annealing temperature resulted in the transmittance's enhancement of the synthesized GdNiSe material. The transmittance of the material decreases with increasing annealing temperatures, leading to improved photoluminescence properties. The material's transmittance, determined by factors such as wavelength, film thickness, and temperature, refers to the fraction of incident light that can pass through. GdNiSe has been found to have excellent optical transmittance in the visible and near-infrared regions. The crystal and electronic structures of GdNiSe affect its optical properties, such as transmittance. GdNiSe can display unique properties based on temperature, showing both metallic and insulating behavior. GdNiSe shows promise for optical filters, infrared detectors, and thermoelectric devices due to its distinct optical and electronic properties. The material's capacity to transition between metallic and insulating states adds to its appeal for novel electronic and energy-related applications. Figure 2c illustrates the reflectance of GdNiSe material that was hydrothermally synthesized. The reflectance decreases under the wavelength of the light radiation, displaying noticeable peaks in the visible part of the spectra. The absorbance of the synthesized GdNiSe material

was enhanced by increasing the annealing temperature. The reflectance of the material increases as the annealing temperature rises, improving the photoluminescence properties of the synthesized material. The reflectance of the hydrothermally synthesized GdNiSe material is due to its electronic structure. GdNiSe is considered as a narrow-bandgap semiconductor due to its bandgap of around 1.58 eV. The movement of electrons between bands results in the reflection of light in visible and near-infrared wavelengths. The electromagnetic and dielectric properties of the GdNiSe material are enhanced through annealing, leading to increased AC conductivity and dielectric constant compared to the unannealed GdNiSe. Annealing the material results in enhanced reflectance. Annealing enhanced the structural and optical characteristics of GdNiSe materials. The reflectance edge shifted slightly toward lower wavelengths due to the material annealing. Changes in the reflectance spectra of GdNiSe material occur when annealing temperature is varied from 220 to 260 °C.

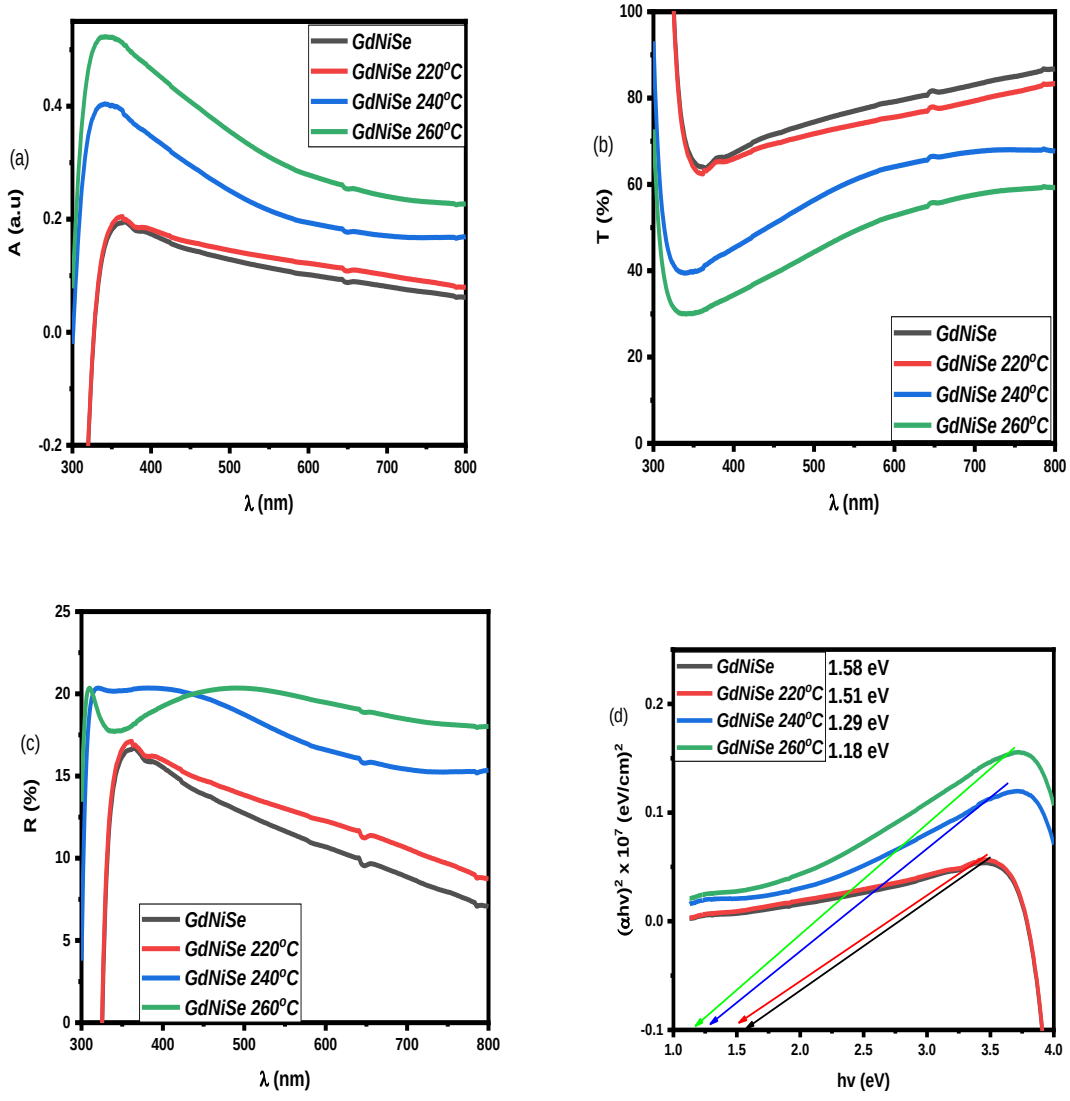


Figure 2: (a) GdNiSe-absorbance, (b) GdNiSe-transmittance, (c) GdNiSe-reflectance, and (d) GdNiSe-bandgap energy

As the annealing temperature increases, the material's reflectance edge shifts toward lower energies, showing a narrower bandgap. Structural changes occurred due to the annealing process, enhancing the crystallinity of the GdNiSe layer. The annealing temperature affects the optical properties of GdNiSe. By raising the annealing temperatures, the material's reflectance spectrum changes as crystallinity improves and defects decrease. GdNiSe material can be advantageous for optoelectronic devices such as solar cells and photodetectors. GdNiSe's high reflectivity in the visible and near-infrared regions makes it perfect for light-harvesting uses. The bandgap separates the valence band and the conduction band. To enable the electron conduction in the conduction band, a minimum energy known as the bandgap is necessary. The only way for electrons to become free is by inputting energy, which bridges the gap between the valence band and the higher energy conduction band. The relationship between $(\alpha h\nu)^2$ and $h\nu$ is shown in Figure 2d. The energy bandgaps for unannealed were approximated by extending the linear portion of the graph to $\alpha h\nu = 0$. The materials' energy bandgaps ranged from 1.58 eV to 1.18 eV. Annealing the material led to a reduction in the energy bandgap. The energy bandgap of GdNiSe was enhanced through the annealing process . The bandgap energy range found in this study is ideal for the absorber layer in solar cells. It absorbs solar energy radiation above 1.18 eV and aligns well with the sun's spectrum (bandgap ~1.5eV).

The refractive index of GdNiSe material varies with photon energy, as depicted in Figure 3a. The bending of light passing through the material varies with the wavelength of the light. The refractive index of GdNiSe material plays a crucial role in optics, photonics, and optoelectronics applications. It is utilized for the creation and production of optical devices such as lenses, prisms, and waveguides.The refractive index of synthesized GdNiSe material is impacted by varying annealing temperatures (220-260 °C). The annealing temperature has a significant impact on the refractive index of GdNiSe. The refractive index rises with higher annealing temperatures. The refractive index plays a vital role in determining how light interacts with a material. It affects aspects such as reflection, refraction, and dispersion. The extinction coefficient of GdNiSe material rises as the post-deposition annealing temperature increases, as depicted in Figure 3b. The absorption edge (Urbach region) increases, resulting in this effect. The extinction coefficient of GdNiSe material increases as the annealing temperature rises from (220-260 °C). The extinction coefficient of GdNiSe increases with higher annealing temperatures. The reason behind this is the enhancement of crystallinity and the decrease in material defects. The extinction coefficient quantifies how well a material absorbs and scatters light. The extinction coefficient is higher when there is more light loss from absorption and scattering. Optical conductivity allows for the measurement of a material's capability to conduct electrical current under light exposure. GdNiSe shows high optical conductivity, making it an ideal choice for optoelectronic devices. GdNiSe possesses a distinctive crystal structure that grants it a significant optical conductivity due to its layered nature. Its characteristics make it suitable for various uses, such as solar cells, LEDs, and photodetectors. GdNiSe is being studied for various applications, such as solar cells and light-emitting diodes, due to its high optical conductivity. Next-generation electronic devices are also being researched for its use. The optical conductivity of GdNiSe material varies significantly with photon energy when synthesized at different annealing temperatures. Inter-band transitions dominate the optical conductivity at low photon energies, while intra-band transitions dominate at higher photon energies. The optical conductivity of GdNiSe is affected by the annealing temperature, as depicted in Figure 3c. With an increase in annealing temperature, the optical conductivity rises, suggesting a rise in the number of free carriers. The structural and optical properties of GdNiSe are closely linked to its optical conductivity. The optical conductivity is influenced by the bandgap, Urbach energy, and dispersion of the material. The dielectric constant of GdNiSe material represents its capacity to store electrical energy under an external electric field, as depicted in Figure 3(d,e).

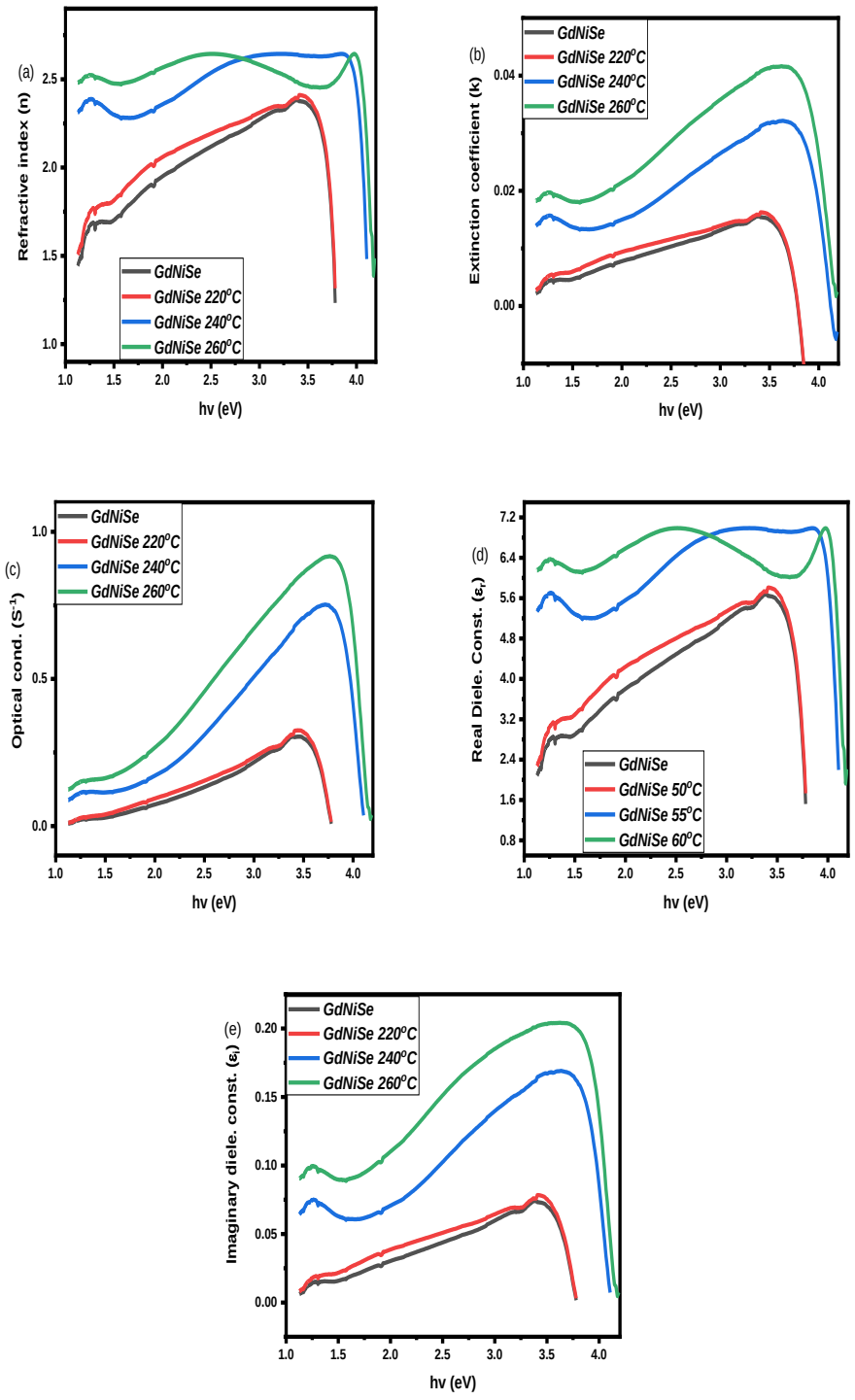


Figure 3: (a) GdNiSe-refractive index, (b) GdNiSe-extinction coefficient, (c) GdNiSe-optical conductivity, and (d and e) GdNiSe-real and imaginary dielectric constant

It signifies the ability of the material to decrease the strength of the electric field within it. The dielectric constant of GdNiSe material is related to its energy dissipation when exposed to an alternating electric field. It shows how a material can transform electrical energy into different

forms. The investigation examined the real and imaginary dielectric constant of GdNiSe material at various annealing temperatures ranging from 220 to 260°C. The results indicate that as the annealing temperature increases, the real dielectric constant increases and the imaginary dielectric constant increases. The changes in the material's microstructure and defect structure are responsible for this behavior.

Photoluminescence Study of GdNiSe Material

The emission of light by a material after absorbing photons is known as photoluminescence. The PL properties of GdNiSe have been the subject of study, with temperature and composition being important influencing factors. The layered structure of GdNiSe results in strong PL in the near-infrared region. Some applications such as LEDs and lasers can benefit from this promising material. The GdNiSe performance was optimized for various applications by studying its photoluminescence properties. This involves examining how various growth conditions, doping, and device structures impact the efficiency of photoluminescence and the wavelength of emission. Photoluminescence analysis is a technique used to characterize the optical properties of materials. In GdNiSe, this analysis involves studying the emission of light by GdNiSe when it is exposed to a light source. The annealing temperature affecta the photoluminescence properties of GdNiSe, as depicted in Figure 4. The annealing temperature influences the structural and optical properties of materials. With GdNiSe, different annealing temperatures lead to variations in its photoluminescence behavior. This is because the annealing process affect the crystal structure, and defects state of GdNiSe. Photoluminescence analysis is widely used in materials science and engineering to characterize the optical properties of materials. It can provide insights into the electronic structure, defect states, and surface properties of materials. The photoluminescence was carried out by exposing the materials to sunlight at different time intervals. The unannealed GdNiSe material was exposed to sunlight at 20-minute intervals, ranging from 0 to 140 minutes. The prominent peaks were identified within the wavelength range of 612 to 661 nm. The GdNiSe materials were heated between 220 and 260 °C, and then exposed to sunlight for various periods ranging from 0 to 60 minutes. Prominent peaks were observed at wavelengths between 613 nm and 615 nm, as well as between 663 nm and 665 nm. The synthesized material is an appropriate choice for photovoltaic applications. The PL emission spectra were gathered for two distinct excitation wavelengths: 612-661 nm and 613-663 nm. A distinct excitonic peak was observed in the UV region for all materials, showing a slight red shift with increasing temperature. The UV PL emission peak of NiSe corresponds to the band edge emission, and it is red shifted from 391 nm to 398 nm as the annealing temperature increases. The increase in gadolinium particle and higher annealing temperature may cause the observed red shift in the PL emission spectra. The stress induced during annealing can also have a significant impact on the shift of the PL emission peak. The observation of band-edge UV emission and visible PL emission bands related to defects has also been made. Improved crystallinity in the material can cause PL emission intensity. Annealing the as-synthesized material at a higher temperature of 260 °C resulted in improved crystallinity.

Structural Study of GdNiSe Material

Figure 5 displays the results of using XRD to investigate the structure of GdNiSe material. The diffraction patterns of GdNiSe's hexagonal crystal structure closely match JCPDS:01-06-8823. The peaks observed at 2θ angles of 23.076°, 27.621°, 32.496°, and 36.185° can be accurately assigned to the (111), (112), (116), and (211) diffraction planes [16–20]. The crystal structure and other structural properties of the synthesized materials were determined using XRD analysis. XRD analysis of GdNiSe material has shown that it crystallizes in a hexagonal crystal structure with space group P63/mmc. The lattice parameters of GdNiSe are a = 0.5875 nm and c = 1.0256 nm. The XRD patterns of GdNiSe material are affected by annealing. It impacts the intensity, position, and size of peaks, revealing details about the material's structure and microstructure. The XRD patterns of GdNiSe

showed changes, including increased peak intensities, after annealing, suggesting enhanced crystallinity. Shifting peak positions during annealing implies alterations in lattice parameters and crystal structure [21–24]. Analyzing the XRD patterns pre- and post-annealing provides insights into GdNiSe's microstructural evolution. Annealing impacts the growth of grains, reduction of defects, and transformations in phases, altering the material's physical and chemical characteristics.

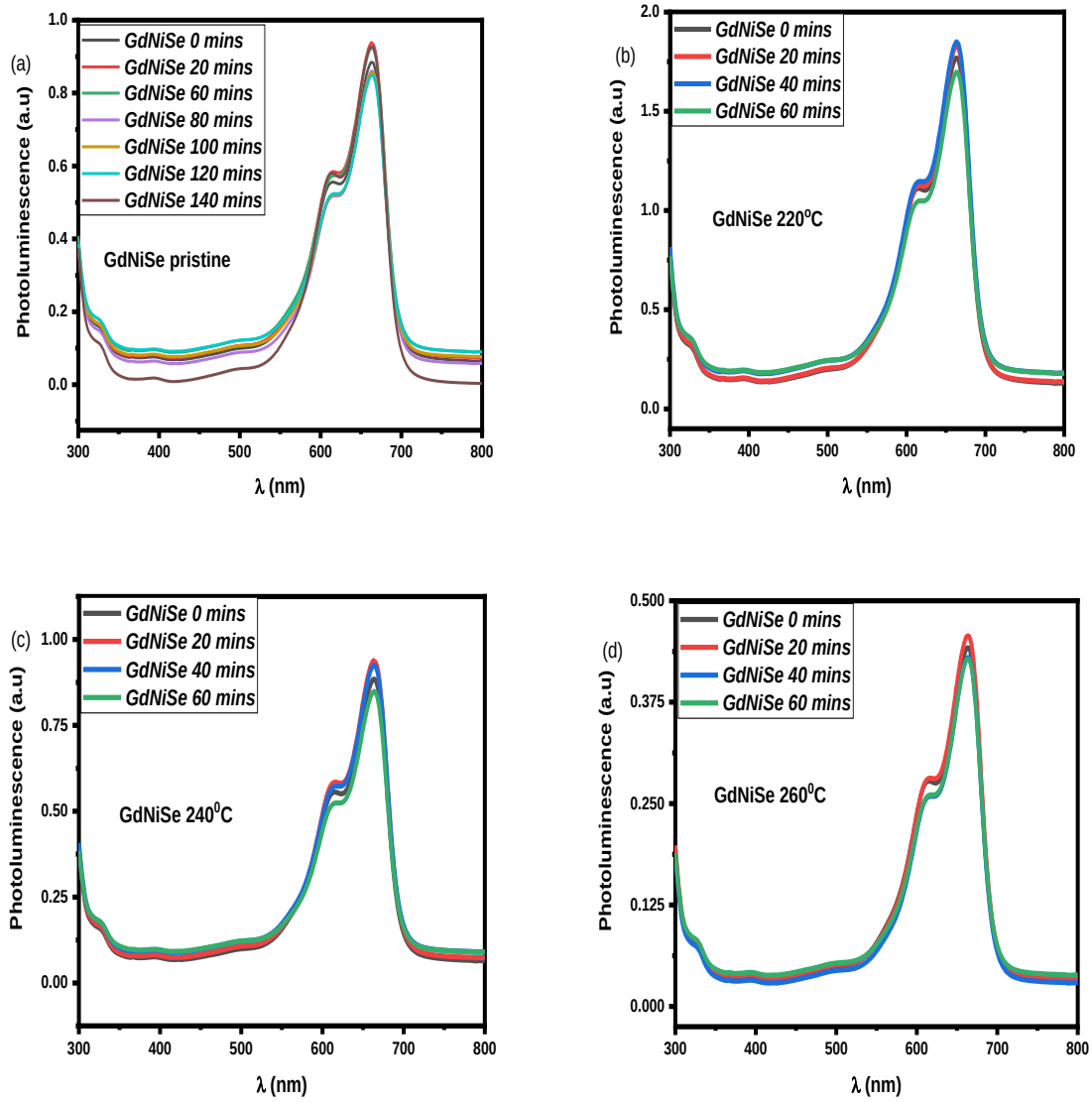


Figure 4: (a) GdNiSe, (b) GdNiSe-220 °C, (c) GdNiSe-240 °C, and (d) GdNiSe-260 °C

Table 1 indicates the structural and microstructural parameters for GdNiSe, derived from XRD analysis. The data reflects a high degree of crystallinity and structural uniformity in the sample. The peaks at 2θ angles of 23.076°, 27.621°, 32.496°, and 36.185° are indexed to the (111), (112), (116), and (211) planes. The indexing and the corresponding lattice parameters ($a \neq c$) confirm that the material crystallizes in a tetragonal crystal system. The calculated lattice parameters are consistently $a = 0.5875$ nm and $c = 1.0256$ nm across all observed peaks. This consistency indicates a stable and well-defined unit cell throughout the material. The crystallite size, determined using the Scherrer equation, ranges from 175.18 nm to

177.81 nm. The average crystallite size is approximately 176.26 nm. Values in this range suggest that the material consists of large nanocrystalline grains, which often leads to high electrical conductivity and distinct magnetic properties compared to smaller nanoparticles. The dislocation density remains relatively constant at approximately $3.2 \times 10^{-5}\text{nm}^{-2}$. The microstrain values range from 6.33×10^{-4} to 9.75×10^{-4}. There is a noticeable decrease in microstrain as the diffraction angle (2θ) increases. This suggests that the internal stress or lattice deformation is slightly more pronounced at the lower-index planes (111) compared to the higher-index planes. The overall low magnitude of strain signifies a relaxed lattice with little internal pressure or distortion. The d-spacing decreases from 0.3850 nm to 0.2480 nm as the 2θ angle increases [25-29]. This follows Bragg's Law, confirming the accuracy of the peak assignments to their respective Miller indices (hkl).

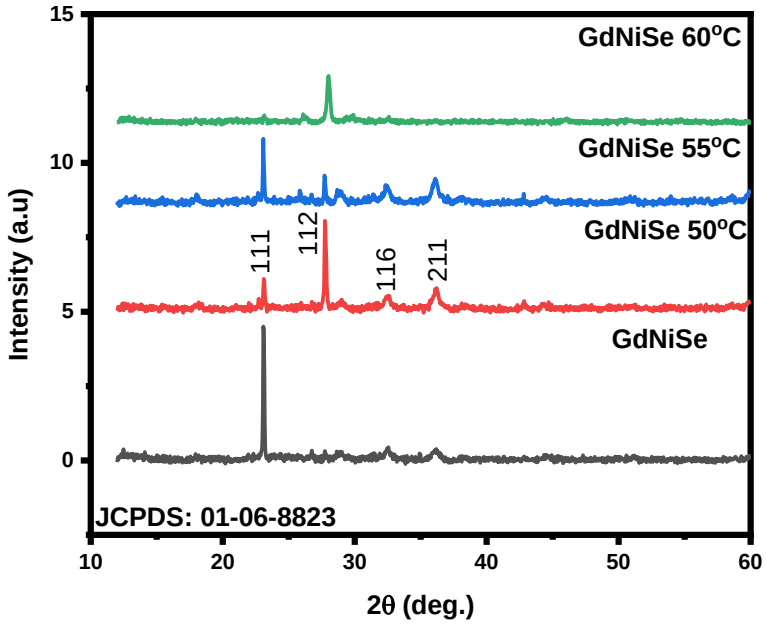


Figure 5: XRD pattern of GdNiSe material

Table 1: Structural parameters of GdNiSe material

2θ (∘)	(hkl)	d -spacing (nm)	a (nm)	c (nm)	D (nm)	δ ($\times 10^{-5}\text{nm}^{-2}$)	ϵ ($\times 10^{-4}$)
23.076	(111)	0.3850	0.5875	1.0256	177.81	3.16	9.75
27.621	(112)	0.3227	0.5875	1.0256	175.18	3.26	8.31
32.496	(116)	0.2753	0.5875	1.0256	176.44	3.21	7.04
36.185	(211)	0.2480	0.5875	1.0256	175.59	3.24	6.33

FTIR Study of GdNiSe Material

FTIR analysis of GdNiSe synthesized at different annealing temperatures (220-260 °C) reveals changes in the functional groups and bonding characteristics of the material, as depicted in Figure 6. By studying GdNiSe using FTIR, the material's vibrational modes are uncovered, shedding light on its structural and electronic properties. This study recognizes the distinct peaks that represent various functional groups and bonds in the material. Infrared radiation is used to investigate the molecular vibrations of the GdNiSe. Vibrational frequencies and intensities of functional groups are determined

by measuring the absorption and transmission of infrared radiation. The crystal structure of GdNiSe is hexagonal, and it is a layered material. The $[GdNiSe]^{2-}$ anion in GdNiSe comprises a gadolinium ion (Gd^{3+}) surrounded by two nickel ions (Ni^{2+}) and two selenium ions (Se^{2-}). The hexagonal crystal structure of NiSe classifies it as a layered transition metal dichalcogenide material. NiSe has a covalent bond between a nickel atom and a selenium atom, known as the Ni-Se bond. The FTIR spectra showed various peaks, such as 2,654, 1,869, 1,638, 1,105, and 935 cm^{-1}. GdNiSe material exhibits a strong covalent bond, leading to a short bond length and high bond strength. The overlapping of their 3d and 4p orbitals creates the bond between Ni and Se, resulting in a bond order of two. The formation of GdNiSe occur in the far-infrared and lower mid-infrared regions (700 cm^{-1}). These bands correspond to the stretching and bending modes of Gd–Se and Ni–Se bonds. A broad absorption band appears between 3,200 and 3,600 cm^{-1}, with a corresponding bending vibration around 1,600–1,640 cm^{-1}. These peaks are attributed to stretching vibrations of water molecules or hydroxyl groups adsorbed on the surface of the nanoparticles. Peaks near 2,360 cm^{-1} (atmospheric CO_2) and between 1,300 and 1,500 cm^{-1} are observed due to the adsorption of atmospheric gases or residual organic precursors used during chemical synthesis. Characteristic peaks for selenium-containing compounds appear in the fingerprint region (500–1,200 cm^{-1}), indicating structural changes and specific phase formations.

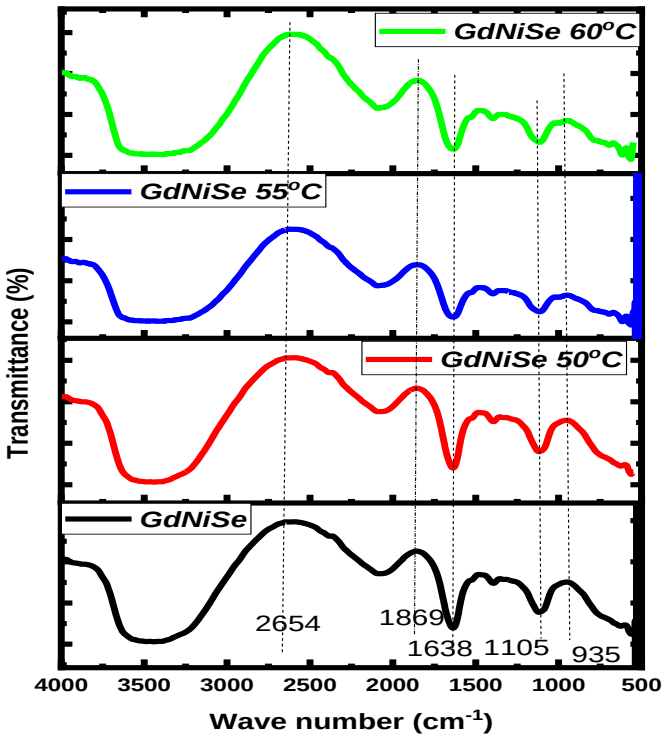


Figure 6: FTIR spectrum of GdNiSe material

Conclusion

The hydrothermal approach was successful in synthesizing GdNiSe. The range of energy bandgaps for the materials was 1.58 eV to 1.18 eV. The energy bandgap decreased as the material was annealed. Annealing decreased the energy bandgap of GdNiSe. The materials were exposed to sunlight at various time intervals for photoluminescence testing. The unannealed GdNiSe material was subjected to sunlight for different periods, ranging from 0 to 140 minutes at 20-minute intervals. Peaks that stood out were found between 612 and 661 nm. The XRD

analysis of GdNiSe material revealed a hexagonal crystal structure with space group P63/mmc. GdNiSe has lattice parameters of a = 0.5875 nm and c = 1.0256 nm. Annealing impacts the XRD patterns of GdNiSe material. It affects the strength, location, and dimensions of peaks, exposing information about the material's composition and arrangement. The FTIR spectra displayed several peaks, including 2,654, 1,869, 1,638, 1,105, and 935 cm^{-1}. The GdNiSe material has a strong covalent bond, resulting in a short bond length and high bond strength. The bond between Ni and Se is formed through the overlap of their 3d and 4p orbitals, resulting in a bond order of two.

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